Wood Design

Report 01

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Table of Contents

Abstract 1

1 Tension Test (*Essai de Traction*) 2

1.1 Introduction 2

1.2 Objective 2

1.3 Calculations and Results 2

1.4 Conclusion 2

2 Axial Compression Test (*Essai de Compression Axiale*) 3

2.1 Introduction 3

2.2 Objective 3

2.3 Calculations and Results 3

2.4 Conclusion 3

3 Local Compression Test (*Essai de Compression Locale*) 4

3.1 Introduction 4

3.2 Objective 4

3.3 Calculations and Results 4

3.4 Conclusion 4

4 Shear Test (*Essai de Cisaillement*) 5

4.1 Introduction 5

4.2 Objective 5

4.3 Calculations and Results 5

4.3.1 Sample Dimensions 5

4.3.2 Machine Test Results 6

4.3.3 Shear Stress Calculation 6

4.3.4 Statistical Analysis 7

4.3.5 Characteristic Shear resistance 8

4.3.5.1 Method 1- Wood Handbook Approach . 8

4.3.5.2 Method 2 - 5th percentile Method 8

4.3.5.3 Method 3 -Euripean Standard EN 384 . 9

4.4 Conclusion 9

5 Bending Test (*Essai de Flexion*) 10

5.1 Introduction 10
5.2 Objective 10
5.3 Calculations and Results 10
5.4 Conclusion 10
General Conclusion 11
References 12
Appendices 13

List of Figures

Figure 4.1 Shear testing's sample 6

List of Tables

Table 4.1. Dimension of shear testing sample 6
Table 4.2. Machine test results 6

Abstract

This is our teams abstrace we may talk about Brief overview of objectives, methodology, and key findings at this section.

1. Tension Test (*Essai de Traction*)

1.1 Introduction

1.2 Objective

1.3 Calculations and Results

1.4 Conclusion

2. Axial Compression Test (*Essai de Compression Axiale*)

2.1 Introduction

2.2 Objective

2.3 Calculations and Results

2.4 Conclusion

3. Local Compression Test (*Essai de Compression Locale*)

3.1 Introduction

3.2 Objective

3.3 Calculations and Results

3.4 Conclusion

4. Shear Test (*Essai de Cisaillement*)

4.1 Introduction

Shear stress in timber occurs when forces act parallel to the grain direction, causing one layer of fibers to slide over another along a plane parallel to the grain. This type of failure is particularly critical in notched beams, connections, and lap joints.

In timber structures, shear failure (cisaillement) is one of the most sudden and brittle failure modes, as wood has significantly lower shear strength parallel to grain compared to its tensile or compressive strength. Understanding shear behaviour is therefore essential for safe structural design.

The design shear stress must satisfy:

$$\frac{\tau_d}{f_{v,d}} \leq 1$$

If the beam has a notch (entaille) at the support:

$$\frac{\tau_d}{k_v \cdot f_{v,d}} \leq 1$$

4.2 Objective

The objective of this shear test are as follows:

1. To determine the shear stress ($\tau_{v,d}$) of each timber specimen parallet to the grain.
2. To compute the mean shear stress (τ_m), standard deviation ($\bar{S}$), and coefficient of variation (C_v).
3. To determine the characteristic shear resistance $f_{v,k}$ using Method 1 (Wood Handbook): $\sigma_{b12,v} = \tau_m - S_m$, then $f_{v,k} = \sigma_{b12,m} \times p_k$ (for shear $p_k = 14\%$).
4. To verify the result using formula: $f_{v,k} = 0.2 f_{m,k}^{0.8}$.
5. To observe the shear failure mode and compare experimental result with standard design values.

4.3 Calculations and Results

4.3.1 Sample Dimensions

Three timber lap joint sample were tested on 26 February 2026 using the Machine at ITC lab. The samples are the stepped lap joint geometry. The shear plane area A is defined as the width b multiplied by the shear height h'.

No.	$a(mm)$	$b(mm)$	$h(mm)$	$h' + h(mm)$
1	20.03	29.95	16.18	28.03
2	20.18	30.66	16.65	28.14
3	20.09	29.92	16.29	28.00

Table 4.1: Dimension of shear testing sample

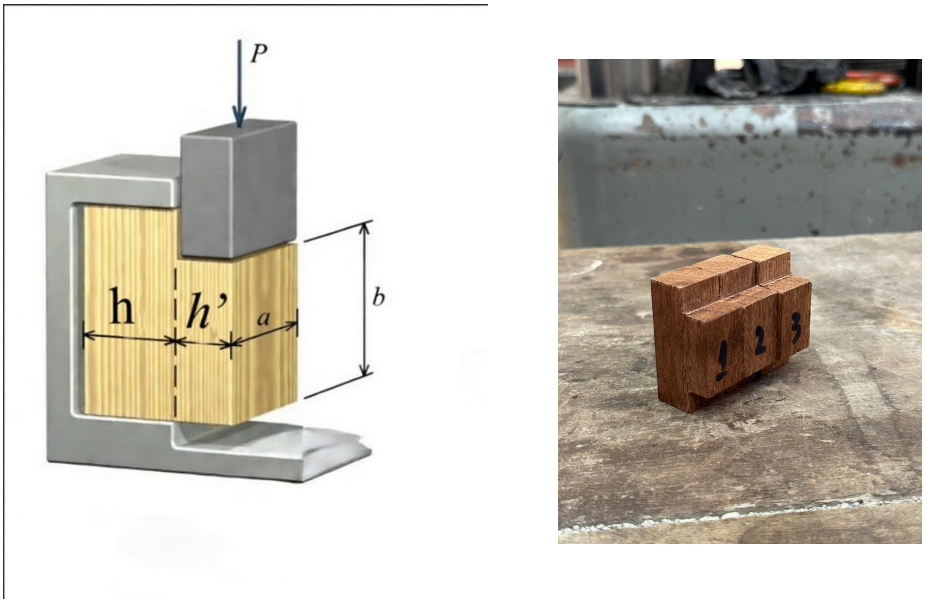

Figure 4.1: Shear testing's sample

4.3.2 Machine Test Results

The following results were recoreded directly from the testing machine.

Test No.	ChNo	a (mm)	b (mm)	S (mm²)	Fm (kN)	Rm (MPa)
5215	N° 1	20.03	29.95	599.9	9.933	16.56
5216	N° 2	20.18	30.66	618.7	8.513	13.76
5217	N° 3	20.09	29.92	601.1	7.707	12.82
Mean	—	20.10	30.18	606.6	8.718	14.38

Table 4.2: Machine test results

4.3.3 Shear Stress Calculation

For each test piece, we calculate the shear stress at failure (rupture) using the formula:

$$\tau_{v,d} = \frac{F_m}{A} = \frac{F_m}{a \cdot b}$$

where F_m is the maximum applied load (N).

Sample 1 (Test 5215):

$$A_1 = b \times a = 29.95 \times 20.03 = 599.9mm^2$$
$$\tau_1 = \frac{9.933[kN]}{599.9[mm^2]} = 16.558MPa$$

Sample 2 (Test 5216):

$$A_1 = b \times a = 30.66 \times 20.18 = 618.719mm^2$$
$$\tau_2 = \frac{8.513[kN]}{618.719[mm^2]} = 13.759MPa$$

Sample 3 (Test 5217):

$$A_1 = b \times a = 29.92 \times 20.09 = 601.093mm^2$$
$$\tau_2 = \frac{7.707[kN]}{601.093[mm^2]} = 12.822MPa$$

4.3.4 Statistical Analysis

Mean shear stress:

$$\tau_m = \frac{\sum_{i=1}^{n} \tau_i}{n}, \quad (n = 3)$$

then,

$$\tau_m = \frac{16.558 + 13.759 + 12.822}{3} = \boxed{14.38MPa}$$

Standard deviation:

$$\bar{S} = \sqrt{\sum_{i=1}^{n} \frac{(\tau_i - \tau_m)^2)}{n-1}}, \quad (n = 3)$$

then,

$$\bar{S} = \sqrt{\frac{(16.558 - 14.38)^2 + (13.759 - 14.38)^2 + (12.822 - 14.38)^2}{3-1}} = \boxed{1.944MPa}$$

Coefficient of Variation (C_v):

$$C_v = \frac{\bar{S}}{\tau_m} \times 100\% = \frac{1.944}{14.38} \times 100\% = \boxed{13.518\%}$$

4.3.5 Characteristic Shear resistance

Once the individual shear stresses and their statistical parameters (mean and standard deviation) have been established, the characteristic shear resistance $f_{v,k}$ is determined. In timber engineering, the characteristic resistance represents the 5th percentile value of the strength distribution. It's mean that 95% of samples are expected to be stronger than this value.

Three different methods are applied and compared in this section, and all are based on the same experimental data that we found in section 4.3.4, but we going to use different statistical and normative approaches as mentioned in lecture note in the **ITC Construction en Bois** course:

Method	Standard/References	Approach	$f_{v,k}$
Method 1	Wood Handbook (2010)	$\sigma_{b12,v} = \tau_m - S_m$	$\sigma_{b12} \times p_k$
Method 2	ITC Courese Notes	$R_n = \bar{R} - 1.645\bar{S}$	R_n/k_z
Method 3	EN 373, EN 384	$\bar{f}_{05} = f_m - 1.165\bar{S}$	$f_{05} \times k_s \times k_v$

The theree methods are detailed below.

4.3.5.1 Method 1- Wood Handbook Approach

Adjusted mean stress at 12% moisture contents:

$$\sigma_{b12,v} = \tau_m - S_m = 14.38 - 1.944 = 12.436MPa$$

Characteristic shear resistance:

$$f_{v,k} = \sigma_{b12,v} \times p_k$$

For shear $p_k = 14$, then

$$f_{v,k1} = 12.436 \times 0.14 = \boxed{1.741MPa}$$

4.3.5.2 Method 2 - 5th percentile Method

This method computed the 5th percentile resistance and divides by a safety factor k_z.

5th percentile resistance:

$$R_n = \bar{R} - 1.645 \times \bar{S}$$

$\bar{R} = \tau_m$, then

$$R_n = 14.38 - 1.645 \times 1.944 = 11.182MPa$$

Characteristic resistance:

$$f_{v,k} = \frac{R_n}{k_z}$$

For shear resistance $k_z = 2.25$, then

$$f_{v,k2} = \frac{11.182}{2.25} = \boxed{4.97MPa}$$

4.3.5.3 Method 3 -Euripean Standard EN 384

The method we are using the 5th percentile value adjusted by k_s and k_v.

5th percentile value:

$$f_{05} = f_i - 1.645\bar{S} \Rightarrow \bar{f}_{05} = f_m - 1.645\bar{S}$$

Then,

$$\bar{f}_{05} = 14.38 - 1.645 \times 1.944 = 11.182MPa$$

Since $\bar{f}_{05} = 13.7664MPa < 30MPa$, then $k_v = 1.12$. And choose $k_s = 0.7$

4.4 Conclusion

5. Bending Test (*Essai de Flexion*)

5.1 Introduction

5.2 Objective

5.3 Calculations and Results

5.4 Conclusion

General Conclusion

This part may be add if we think it is needed.

References

[1] Let say it is References 1

[2] Let say it is References 2 ...

Appendices

This is appendices